ANALOG SYSTEMS : PROBLEM SET 1

Problem 1

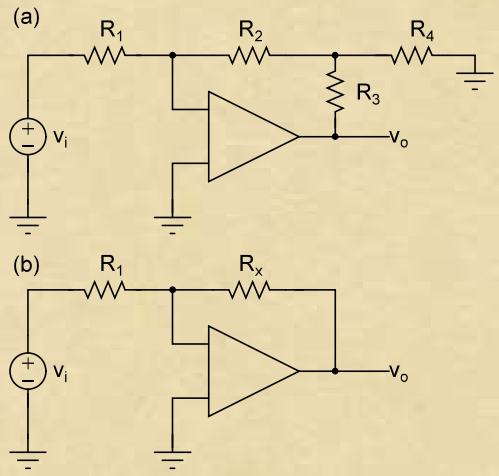


Figure 1: Circuits for Problem 1.

In the circuits above, determine the signs on the opamps for negative feedback operation, and determine v_o/v_i. When realizing a gain with a large magnitude, what might be the advantage of the circuit of (a) over that in (b)?

Problem 2

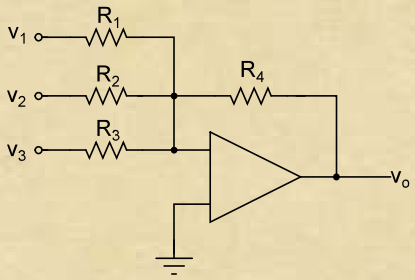


Figure 2: Circuit for Problem 2.

In the circuit above, determine the signs on the opamp to ensure negative feedback. Also determine v_o in terms of v_1, v_2 and v_3.

Problem 3

In the figure, the opamp is operated with dual supplies of ± 10 V. The saturation limits of may be assumed to be ± 9 V. The input to the amplifier is shown. Sketch the output and the voltage v_x at the virtual ground node. What is the maximum input amplitude that will ensure a distortion free output?

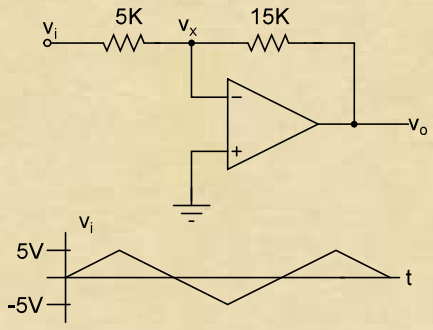


Figure 3: Circuit for Problem 3.

Problem 4

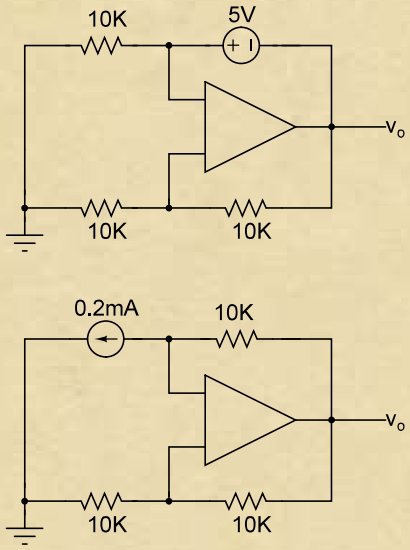


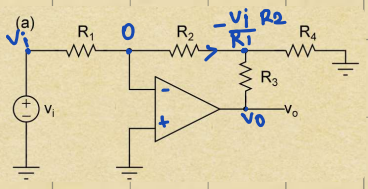
Figure 4: Circuits for Problem 4.

In the circuits above, mark the signs on the opamp for negative feedback operation, and determine v_o.

Problem 5

In the circuits above, mark the signs on the opamp for negative feedback operation, and determine v_o.

(1)



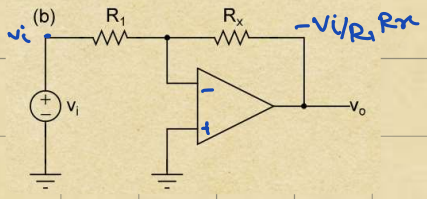
$$\frac{-v_i(R_2/R_1) - v_o}{R_3} + \frac{-v_i(R_2/R_1)}{R_4} = \frac{v_i}{R_1}$$

$$\Rightarrow -v_i(R_2/R_1)\left(1/R_3 + 1/R_4\right) - v_o/R_3 = \frac{v_i}{R_1}$$

$$\Rightarrow -\frac{v_i}{R_1}\left[1 + \frac{R_2}{R_3} + \frac{R_2}{R_4}\right] = \frac{v_o}{R_3}$$

$$\Rightarrow -\frac{v_i}{R_1} \cdot R_2\left[1/R_2 + 1/R_3 + 1/R_4\right] = \frac{v_o}{R_3}$$

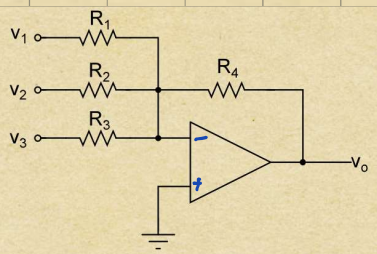
$$\frac{v_o}{v_i} = -\frac{R_3 R_2}{R_1}\left[\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}\right]$$



$$v_o = -v_i/R_1 R_x$$

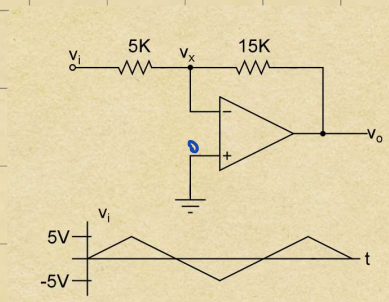
$$\Rightarrow v_o/v_i = -(R_x/R_1)$$

(2)



$$-\left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3}\right) R_4 = v_o$$

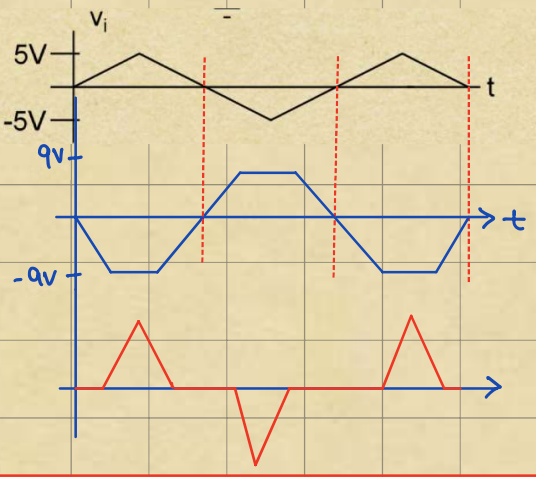
(3)



$$i = \frac{v_i}{5K}$$

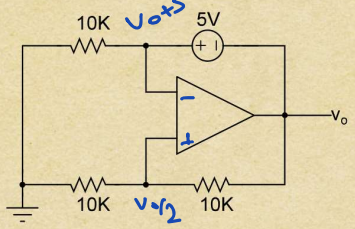
$v_o = -3v_i$

$0x = 0$

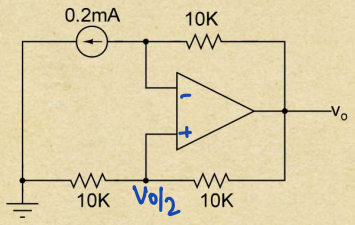


max v_{in} for no distortion is 3V

(4)



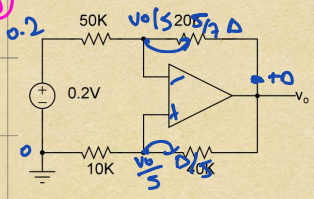
$$\frac{V_o+5}{2} = \frac{V_o}{2} \Rightarrow V_o = -10$$



$$\frac{0-V_o}{2} = 2\times10^{-4}\times10^{4} = 2 \Rightarrow V_o = -4$$

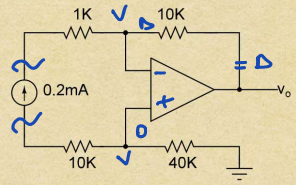
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(5)



$$\frac{V_o - V_o}{5} = i\times 2\times10^{4} \Rightarrow \frac{2}{5}V_o = 2\times10^{4} i \qquad V_o = 25\times10^{4} i$$

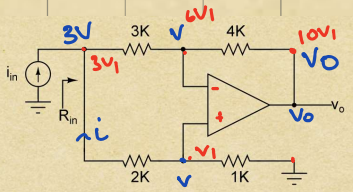
$$\frac{V_o}{5} - 0.2 = \frac{V_o}{25}\times10^{-4}\times\frac{2}{5}\times10^{4} \Rightarrow \frac{9V_o}{5} = -0.2 \Rightarrow V_o = -1/9\,V$$



$$V = -0.2\times10^{-3}\times4\times10^{4} = -8V \qquad V - V_o = 2 \Rightarrow V_o = -10V$$

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(6)



$$V_o = -1000\,i_n$$

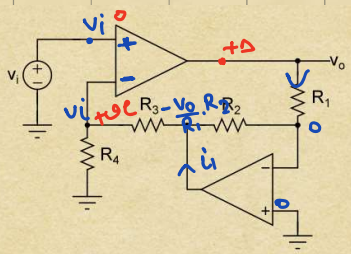
$$i = -V\times10^{-3} \qquad i + i_1 = \frac{V - V_o}{4}\times10^{-3}$$

$$2V = (i_1 + i)\cdot 3\times10^{-3}$$

$$\frac{2V}{3}\times10^{3} = i_1 - V\times10^{3} \Rightarrow i_1 = \frac{5V}{3}\times10^{-3} \qquad R_{in} = \frac{3V\times3\times10^{-3}}{5V} = 1.8\,K\Omega$$

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(8)



$$\frac{V_o}{R_1} \qquad \frac{-V_o}{R_1}\times R_2 \qquad V_i \qquad \frac{V_o}{R_1}$$

$$i = \frac{V_o}{R_1}$$

$$-V_o\left(\frac{R_2}{R_1}\right) - V_i = (i + i_1)R_3 \Rightarrow V_o\left(\frac{R_2+R_3}{R_1}\right) = V_i \Rightarrow V_o = -\frac{R_1 V_i}{R_2+R_3}$$

$$V_i/R_4 = i + i_1 \qquad i_1 = V_i/R_4 - V_o/R_1 \qquad -V_o(R_2/R_1) = V_i(R_3/R_4+1)$$

$$\Rightarrow V_o = -V_i(R_1/R_2)(R_3/R_4+1)$$

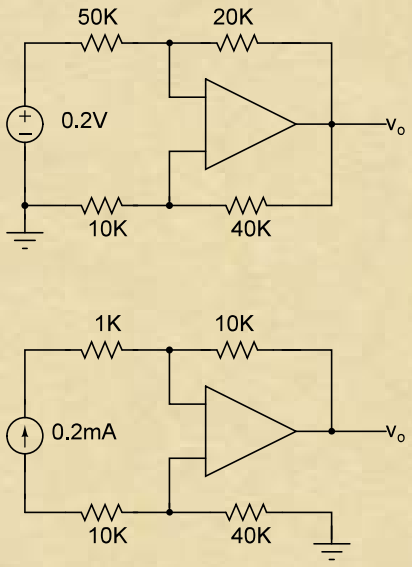


Figure 5: Circuits for Problem 5.

Problem 6

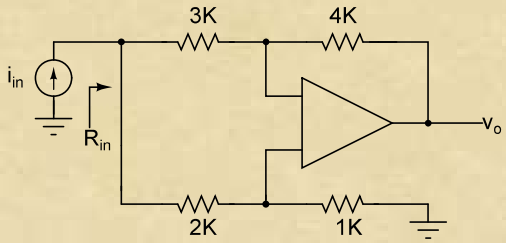


Figure 6: Circuits for Problem 6.

In the circuit above, mark the signs on the opamp for negative feedback operation, and determine v_o. Determine the input resistance looking in, as denoted by R_{in}.

Problem 7

In the circuit above, mark the signs on the opamp for negative feedback operation, and determine v_o in terms of $v_1, \cdots, v_6$.

Problem 8

In the circuit above, mark the signs on the opamp for negative feedback operation, and determine v_o.

Problem 9

In the circuit above, mark the signs on the opamp for negative feedback operation, and determine the Norton equivalent for the circuit looking across the load resistor R_L.

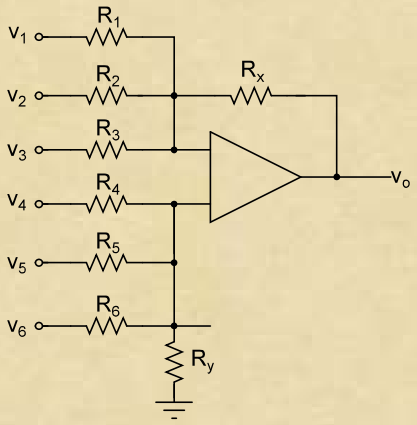


Figure 7: Circuit for Problem 7.

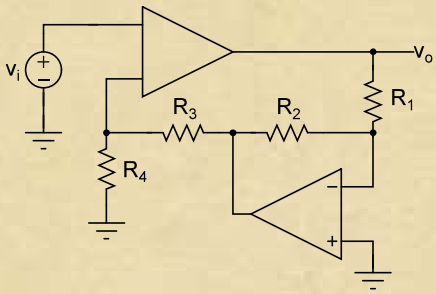


Figure 8: Circuit for Problem 8.

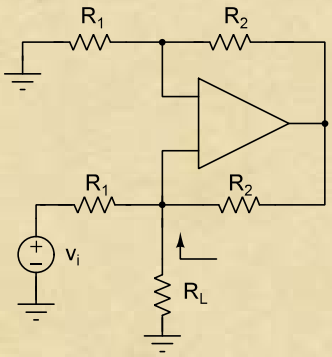
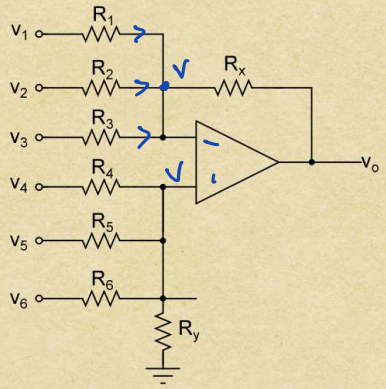


Figure 9: Circuit for Problem 9.

(8)



$$\frac{V_1-V}{R_1}+\frac{V_2-V}{R_2}+\frac{V_3-V}{R_3}=\frac{V-V_o}{R_x}$$

$$\frac{V_4-V}{R_4}+\frac{V_5-V}{R_5}+\frac{V_6-V}{R_6}=\frac{V}{R_y}$$

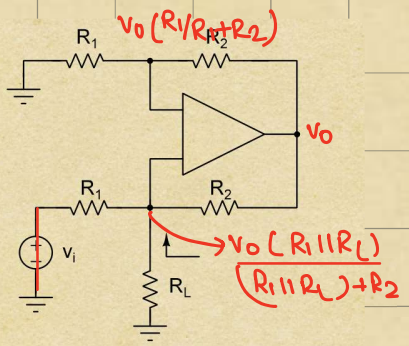
$$\frac{V_1}{R_1}+\frac{V_2}{R_2}+\frac{V_3}{R_3}+\frac{V_o}{R_x}=V\left(\frac{1}{R_x}+\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)$$

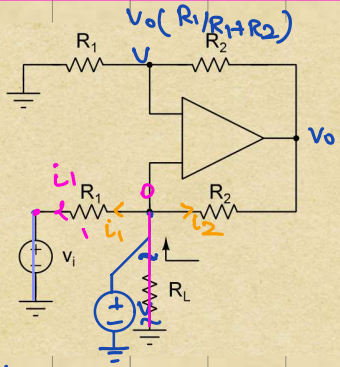
$$\frac{V_4}{R_4}+\frac{V_5}{R_5}+\frac{V_6}{R_6}=V\left(\frac{1}{R_y}+\frac{1}{R_4}+\frac{1}{R_5}+\frac{1}{R_6}\right)$$

$$\frac{\frac{V_1}{R_1}+\frac{V_2}{R_2}+\frac{V_3}{R_3}+\frac{V_o}{R_x}}{\frac{V_4}{R_4}+\frac{V_5}{R_5}+\frac{V_6}{R_6}}=\frac{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_x}}{\frac{1}{R_4}+\frac{1}{R_5}+\frac{1}{R_6}+\frac{1}{R_y}}$$

$$\Rightarrow V_o=R_x\left\{\left[\frac{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_x}}{\frac{1}{R_4}+\frac{1}{R_5}+\frac{1}{R_6}+\frac{1}{R_y}}\right]\left(\frac{V_4}{R_4}+\frac{V_5}{R_5}+\frac{V_6}{R_6}\right)-\left(\frac{V_1}{R_1}+\frac{V_2}{R_2}+\frac{V_3}{R_3}\right)\right\}$$

(9)





$$i_1=-V_i/R_1$$

$$i_2=V/R_1=0$$

$$i_{SC}=\frac{V_i}{R_1}$$

$$V=V_o\left(\frac{R_1}{R_1+R_2}\right) \qquad i_2=\frac{V-V_o}{R_2}=\frac{V}{R_2}\left(-\frac{R_2}{R_1}\right)=-\frac{V}{R_1}$$

$$i_1=V\left(\frac{1}{R_1}\right)\Rightarrow(i_1+i_2)=0\Rightarrow R_{TH}=\frac{V}{0}=\infty$$

$v_e = v_i - \frac{v_o}{n} = v_i - \frac{A v_e}{n} \Rightarrow v_i = v_e\left(1 + \frac{A}{n}\right)$

Problem 10

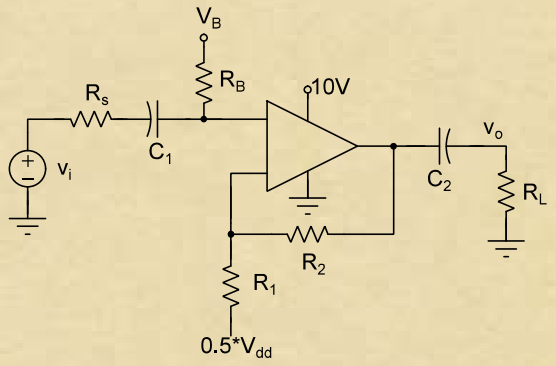


Figure 10: Circuit for Problem 10.

The circuit above shows an amplifier intended for audio applications. The lowest frequency of interest, therefore, is 20 Hz. The opamp is operated with a **single** supply. Assume $V_{DD} = 10\,\text{V}$.

- Mark the signs on the opamp for negative feedback operation.Determine V_B so that no dc current flows through R_1.
- $R_1 = 10\,\text{K}$, $R_s = 50\,\text{K}$ and $R_L = 1\,\text{K}$. $C_1 = 1\,\mu\text{F}$. Determine R_B so that the ac voltage across C_1 at the lowest frequency of interest is less than 1% of v_i.
- Determine C_2 so that the ac voltage across it at the lowest frequency of interest is less than 1% of the ac amplitude across the load resistor.
- Determine R_2 to achieve an ac gain of 50. Sketch the magnitude of the transfer function from v_i to v_o.
- Determine the largest input amplitude of a 1 kHz sinusoid that will result in a distortion free output. The opamp saturates if its output attempts to go to within 1 V of its supply rails.

Problem 11

This problem illustrates another aspect of negative feedback, namely pre-distortion. In the amplifier above, the opamp is non-ideal. We will consider two cases. In the first, the opamp's input characteristic is as shown in Fig. 11(a). Plot the error voltage v_e between the input terminals of the opamp as v_i is swept from $-v_{max}/n$ to v_{max}/n. Assume that $A/n \gg 1$.

In class we assumed that the opamp characteristic saturates abruptly. In reality, saturation occurs in a gentler fashion. An example is shown in Fig. 11(b). On the same graph as you plotted for the previous part, plot v_e as v_i is swept. What do you notice?

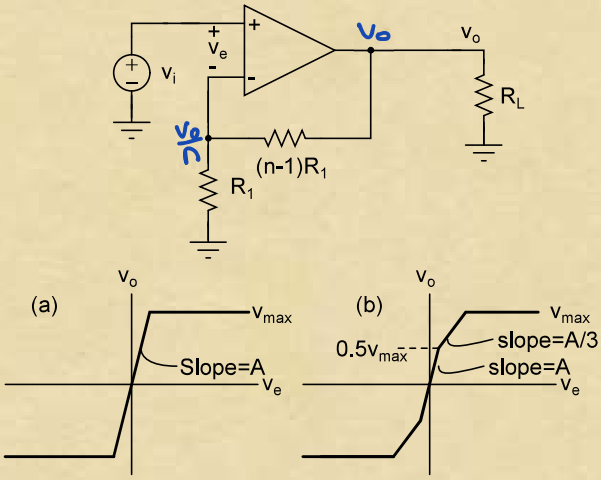


Figure 11: Circuit for Problem 11.

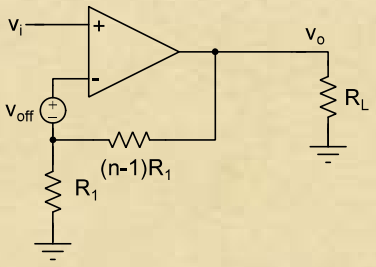


Figure 12: Circuit for Problem 12.

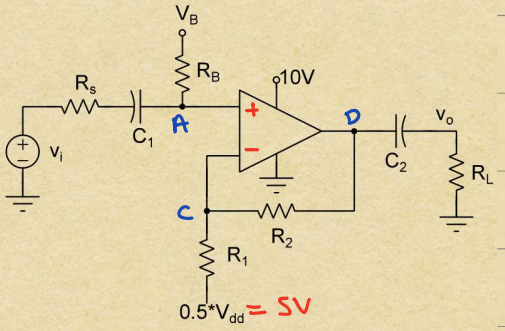
Problem 12

This circuit explores another non-ideality of an opamp, namely **offset**. In an ideal opamp, $v_o = Av_e$, with $A \to \infty$. In reality, it turns out that if the opamp does not saturate, its output can be expressed as $v_o = A(v_e - v_{off})$, with $A \to \infty$. Draw the characteristics of an ideal opamp assuming saturation, and that of the opamp with an offset voltage. What is the output dc offset of the amplifier of Fig. 12?

Problem 13

The figure above shows three different ways of achieving an amplifier with a gain of n^2, where $n^2 \gg 1$. If $v_{off,1,2} = 0$ and the opamps have infinite gain, all three are equivalent. When the opamps have a finite gain A, the gains will deviate from the ideal value of n^2. Determine the output offset voltage and gain in each of the three cases. Make suitable approximations, such as $1/(1+x) \approx 1 - x$ for small x etc. Which of the amplifiers above is least tolerant of finite opamp gain? Which is the most tolerant?

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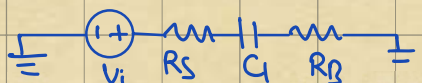


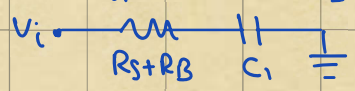
The circuit above shows an amplifier intended for audio applications. The lowest frequency of interest, therefore, is 20 Hz. The opamp is operated with a **single** supply. Assume $V_{DD} = 10$ V.

- Mark the signs on the opamp for negative feedback operation.Determine V_B so that no dc current flows through R_1.
- $R_1 = 10$ K, $R_s = 50$ K and $R_L = 1$ K. $C_1 = 1\,\mu$F. Determine R_B so that the ac voltage across C_1 at the lowest frequency of interest is less than 1% of v_i.
- Determine C_2 so that the ac voltage across it at the lowest frequency of interest is less than 1% of the ac amplitude across the load resistor.
- Determine R_2 to achieve an ac gain of 50. Sketch the magnitude of the transfer function from v_i to v_o.
- Determine the largest input amplitude of a 1 kHz sinusoid that will result in a distortion free output. The opamp saturates if its output attempts to go to within 1 V of its supply rails.

(a) $V_C = 5V \Rightarrow V_A = 5V \Rightarrow V_B = 5V$

(b)





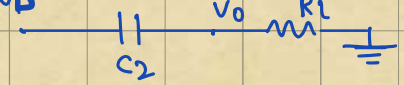
$$\Rightarrow V_{C_1} = \frac{V_i \times \frac{1}{j\omega C_1}}{\frac{1}{j\omega C_1} + (R_S + R_B)} = \frac{V_i}{1 + j\omega C_1 (R_S + R_B)}$$

$$\sqrt{1 + (\omega C_1 (R_S + R_B))^2} \geq 100$$

$$\Rightarrow \omega C_1 (R_S + R_B) \geq 100 \Rightarrow 2\pi \times 20 \times 10^{-6} (R_S + R_B) \geq 100$$

$$\Rightarrow R_S + R_B \geq 795.77\,k\Omega \Rightarrow R_B \geq 745.77\,k\Omega$$

(c)



$$\frac{V_o}{R_L} = \frac{V_D - V_o}{\frac{1}{j\omega C_2}} \Rightarrow \frac{V_D - V_o}{V_D} = \frac{1}{j\omega C_2 R_L}$$

$$\Rightarrow \frac{1}{\omega C_2 R_L} \leq \frac{1}{100} \Rightarrow C_2 \geq \frac{100}{2\pi \times 20 \times 1000} \Rightarrow C_2 \geq 795.77\,\mu F$$

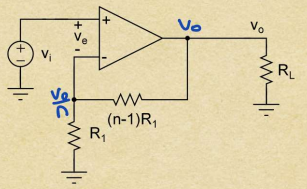
(e) Max = 9, min = 1V $\Rightarrow$ mean = 5V $\Rightarrow$ Amp = 4V (9 − 5)

(d) $\left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_b}{R_b + R_s}\right) = 50 \Rightarrow R_2 = 523.52\,k\Omega$

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$v_e = v_i - \frac{v_o}{n} = v_i - \frac{Av_e}{n} \Rightarrow v_i = v_e\left(1+\frac{A}{n}\right)$

$v_e = \frac{v_i}{1+A/n} \approx v_i\left(\frac{n}{A}\right)$



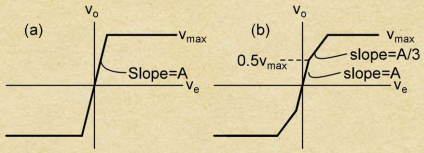
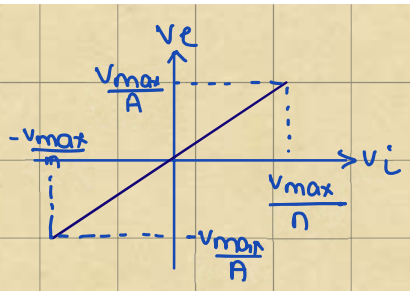


Figure 11: Circuit for Problem 11.

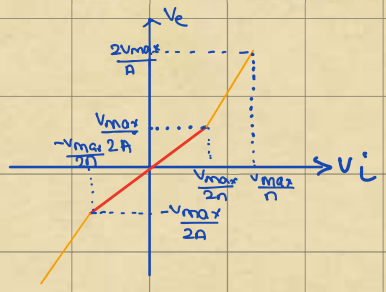
This problem illustrates another aspect of negative feedback, namely pre-distortion. In the amplifier above, the opamp is non-ideal. We will consider two cases. In the first, the opamp's input characteristic is as shown in Fig. 11(a). Plot the error voltage v_e between the input terminals of the opamp as v_i is swept from $-v_{max}/n$ to v_{max}/n. Assume that $A/n \gg 1$.

In class we assumed that the opamp characteristic saturates abruptly. In reality, saturation occurs in a gentler fashion. An example is shown in Fig. 11(b). On the same graph as you plotted for the previous part, plot v_e as v_i is swept. What do you notice?

a)



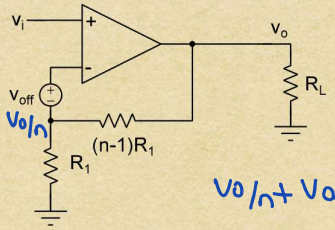
b) $v_e = v_i - \frac{Av_e}{n}$ $\quad v_e < \frac{v_{max}}{2A}$



$v_e = v_i - \frac{v_{max}}{3n} - \frac{Av_e}{3n}$

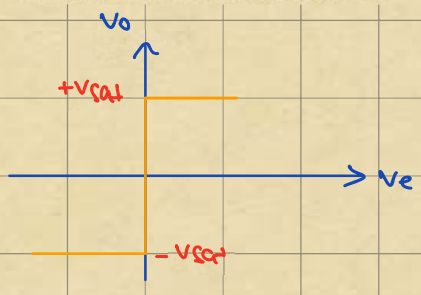
$\frac{Av_e}{3n} = v_i - \frac{v_{max}}{3n}$

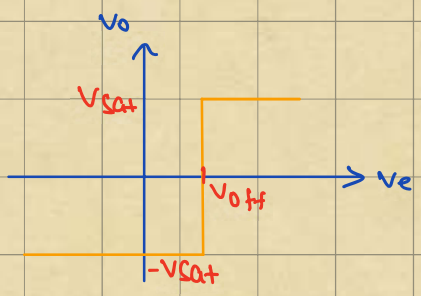
12



This circuit explores another non-ideality of an opamp, namely **offset**. In an ideal opamp, $v_o = Av_e$, with $A \to \infty$. In reality, it turns out that if the opamp does not saturate, its output can be expressed as $v_o = A(v_e - v_{off})$, with $A \to \infty$. Draw the characteristics of an ideal opamp assuming saturation, and that of the opamp with an offset voltage. What is the output dc offset of the amplifier of Fig. 12?

$v_o/n + v_{off} = v_i \Rightarrow v_o = n(v_i - v_{off})$





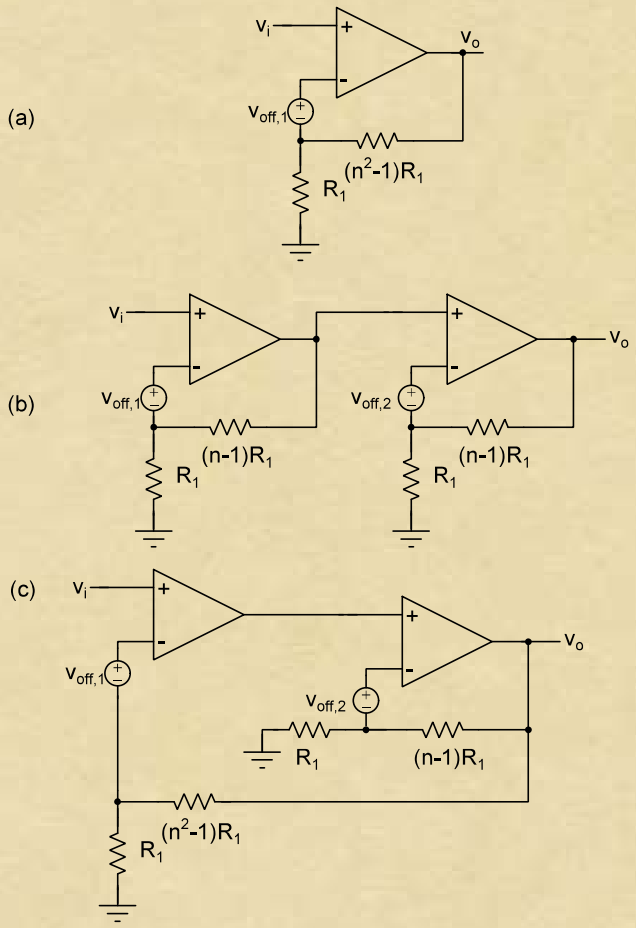


Figure 13: Circuit for Problem 13.

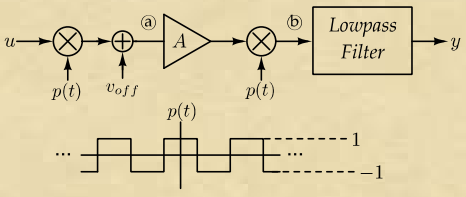


Figure 14: Circuit for Problem 14.

Problem 14

In many precision sensing applications, amplifier offset can be (very) problematic. This problem illustrates the idea of **chopping**, which is one way of solving the offset problem. The amplifier's offset is modeled by v_{off}.The input u is multiplied by a square wave with 50% duty cycle, processed by the amplifier (whose gain is A). The output of the amplifier is multiplied by the same square wave, and passed through a low-pass filter. $p(t)$ has a fundamental frequency of f_c, as shown in Fig. 14. For the purposes of this problem, you can assume that the filter is ideal and has a cut-off frequency smaller than f_c.

Assuming u is dc, plot the signals at ⓐ, the amplifier output, ⓑ and at y. What is y/u?

Repeat the exercise above assuming that the amplifier, in addition to offset, has finite bandwidth. For simplicity, assume that the transfer function of the amplifier is

$$A(s) = \frac{A_o}{1 + s\tau} \tag{1}$$

where $\tau f_c \ll 1$.

Problem 15

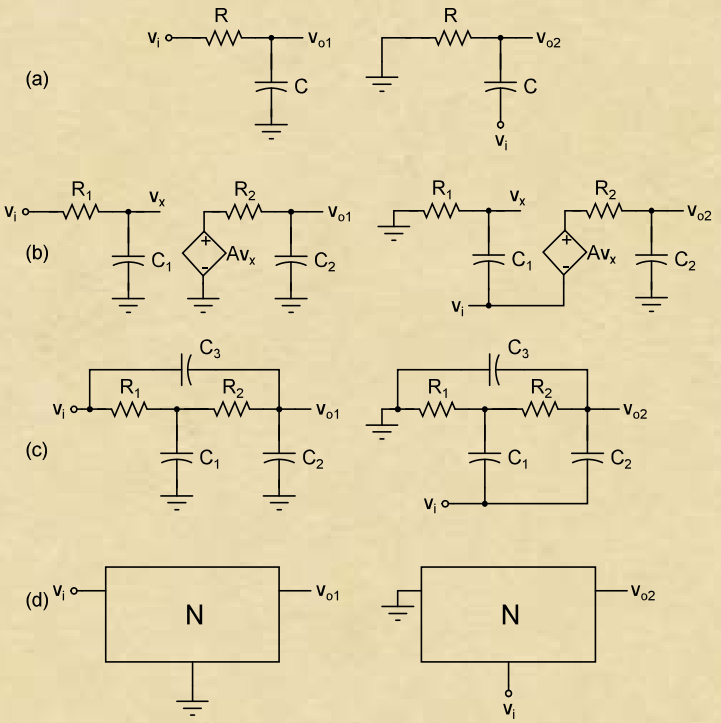
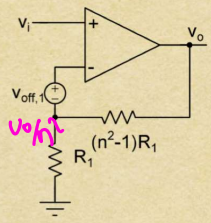


Figure 15: Circuit for Problem 15.

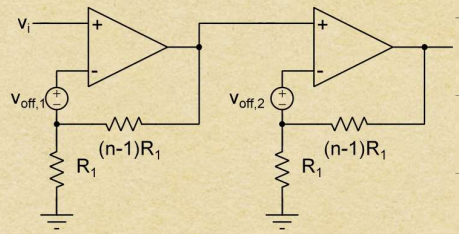
For each of the networks in parts (a), (b) and (c) above, determine $H_1(s) = V_{o1}(s)/V_i(s)$ and $H_2(s) = V_{o2}(s)/V_i(s)$. Is there a pattern you notice? Generalize it to an arbitrary linear network of Fig. 15(d).

(14) The figure above shows three different ways of achieving an amplifier with a gain of n^2, where $n^2 \gg 1$. If $v_{off,1,2} = 0$ and the opamps have infinite gain, all three are equivalent. When the opamps have a finite gain A, the gains will deviate from the ideal value of n^2. Determine the output offset voltage and gain in each of the three cases. Make suitable approximations, such as $1/(1+x) \approx 1-x$ for small x etc. Which of the amplifiers above is least tolerant of finite opamp gain? Which is the most tolerant?

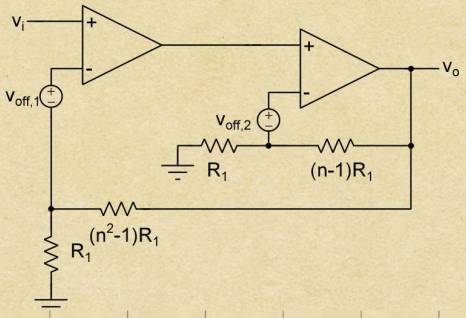
(a)



(b)



(c)



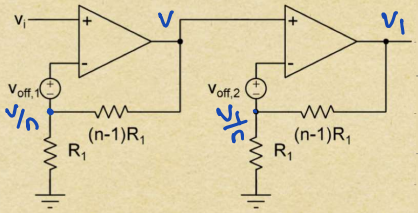
a)

$V_o = A\left(V_i - \frac{V_o}{n^2} - V_{off,1}\right)$ ~~20~~

$V_o\left(1 + \frac{A}{n^2}\right) = AV_i$

$\frac{V_o}{V_i} = \frac{n^2 A}{n^2 + A}$

b)

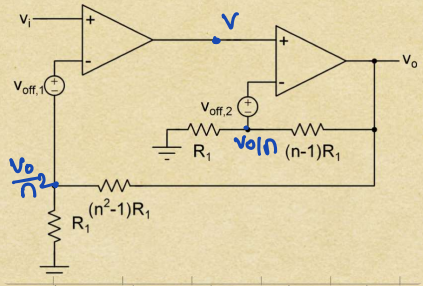


$V = A(V_i - V_1/n) \Rightarrow V(1/A + 1/n) = V_i$

$V_1 = A(V - V_1/n) \Rightarrow V_1(1/A + 1/n) = V_i/(1/A + 1/n)$

$\Rightarrow V_1 = \frac{V_i}{(1/A + 1/n)^2} = \frac{V_i \cdot A^2}{(1 + n/A)^2}$

c)



$V = A\left(V_i - \frac{V_o}{n^2}\right)$

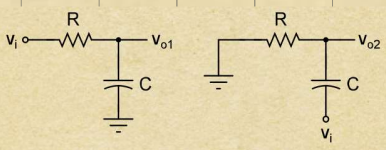
$V_o = A\left(V - \frac{V_o}{n}\right) \Rightarrow V = V_o\left(\frac{1}{A} + \frac{1}{n}\right)$

$\Rightarrow V_o\left(\frac{1}{A} + \frac{1}{n}\right) = A\left(V_i - \frac{V_o}{n^2}\right)$

$\Rightarrow V_o\left(\frac{1}{A^2} + \frac{1}{nA} + \frac{1}{n^2}\right) = V_i \Rightarrow \frac{V_o}{V_i} = \frac{1}{\frac{1}{A^2} + \frac{1}{nA} + \frac{1}{n^2}}$

(15)

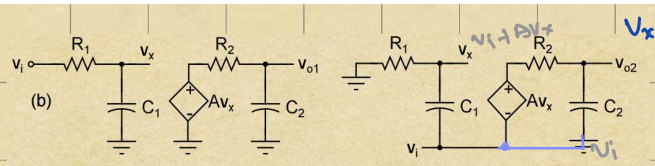
(a)



$\frac{V_{o1}}{V_i} = \frac{\frac{1}{sC}}{\frac{1}{sC} + R} = \frac{1}{1 + sCR}$

$V_{o1} + V_{o2} = V_i$

$\frac{V_{o2}}{V_i} = \frac{R}{\frac{1}{sC} + R} = \frac{sCR}{1 + sCR}$

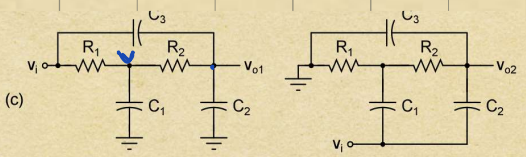


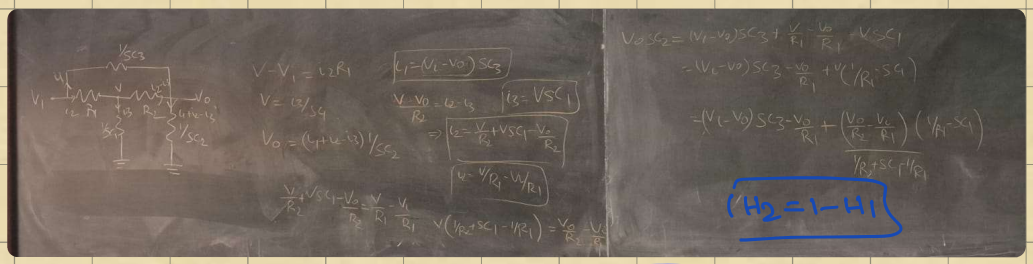
$V_x = \frac{SC_1R_1}{1+SC_1R_1} V_i$

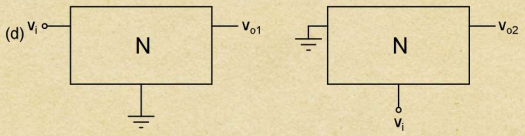
$V_{o1} = AV_x \cdot \frac{\frac{1}{C_2 S}}{\frac{1}{C_2 S} + R_2} = \frac{A}{1+SC_1R_1} \cdot \frac{1}{1+SC_2R_2} V_i$

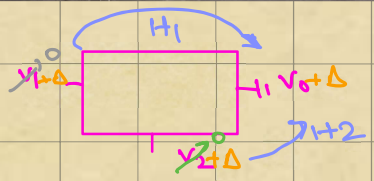
$V_{o2} = \frac{V_i (R_2C_2S)}{1+R_2C_2S} + \frac{V_i + AV_x}{1+R_2C_2S} \Rightarrow V_{o1} + V_{o2} = V_i$

$H_1(S) + H_2(S) = \frac{1}{1+R_2C_2S} \left[V_i R_2C_2S + V_i + A\frac{SC_1R_1}{1+SC_1R_1} V_i + \frac{A V_i}{SC_1R_1+1} \right] = \frac{V_i(2+R_2C_2S)}{1+R_2C_2S}$









$V_2H_2 + V_1H_1 = V_o$

$V_1 = V_2 = V_o = \Delta \Rightarrow H_1 + H_2 = 1$

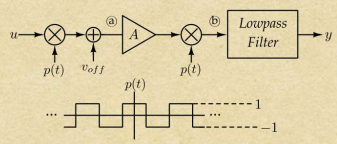


Figure 14: Circuit for Problem 14.

In many precision sensing applications, amplifier offset can be (very) problematic. This problem illustrates the idea of **chopping**, which is one way of solving the offset problem. The amplifier's offset is modeled by v_{off}.The input u is multiplied by a square wave with 50% duty cycle, processed by the amplifier (whose gain is A). The output of the amplifier is multiplied by the same square wave, and passed through a low-pass filter. $p(t)$ has a fundamental frequency of f_c, as shown in Fig. 14. For the purposes of this problem, you can assume that the filter is ideal and has a cut-off frequency smaller than f_c.

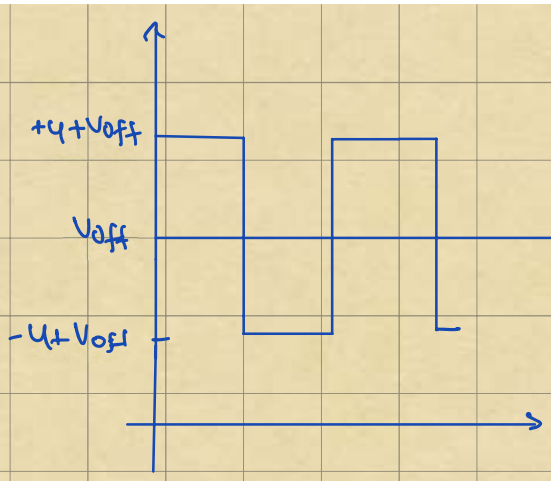
Assuming u is dc, plot the signals at ⓐ, the amplifier output, ⓑ and at y. What is y/u?

Repeat the exercise above assuming that the amplifier, in addition to offset, has finite bandwidth. For simplicity, assume that the transfer function of the amplifier is

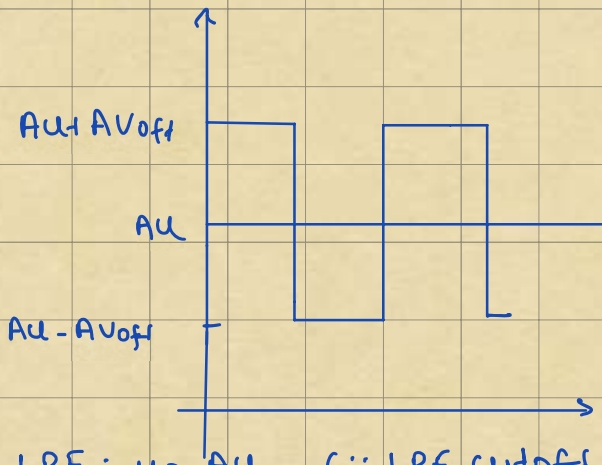
$$A(s) = \frac{A_o}{1 + s\tau} \qquad (1)$$

where $\tau f_c \ll 1$. → BW >> fc

a) $u_P(t)$



b) $V_{out} = A u P(t) + A V_{off} \times P(t) = Au + AV_{off} P(t)$



c) Pass LPF : $y = Au$ ($\because$ LPF cutoff $< f_c$) $\Rightarrow \frac{y}{u} = A$

$$A(s) = \frac{A_o}{1 + s\tau}$$

$(uP(s) + V_{off})\, A(s)$ $\quad P(s) = \frac{1}{s}\left(\frac{1-e^{-sT_c/2}}{1+e^{-sT_c/2}}\right)$

$$\frac{u}{s}\frac{(1-e^{-sT_c/2})}{(1+e^{-sT_c/2})} * \frac{A_o}{1+s\tau} * \frac{u}{s}\frac{(1-e^{-sT_c/2})}{(1+e^{-sT_c/2})}$$

Problem 16

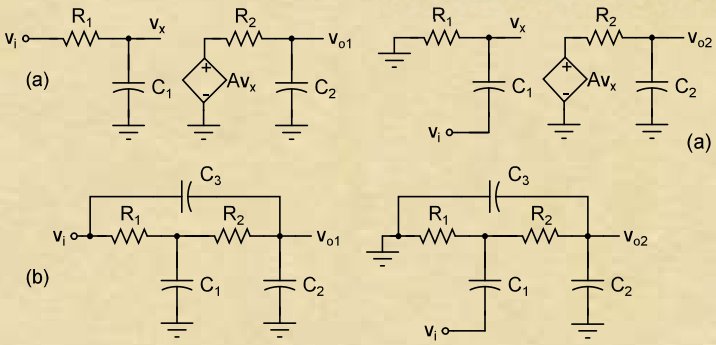


Figure 16: Circuit for Problem 16.

For each of the networks in parts (a), (b) above, determine $H_1(s) = V_{o1}(s)/V_i(s)$ and $H_2(s) = V_{o2}(s)/V_i(s)$. Does the pattern you noticed in the previous problem still hold?

Problem 17

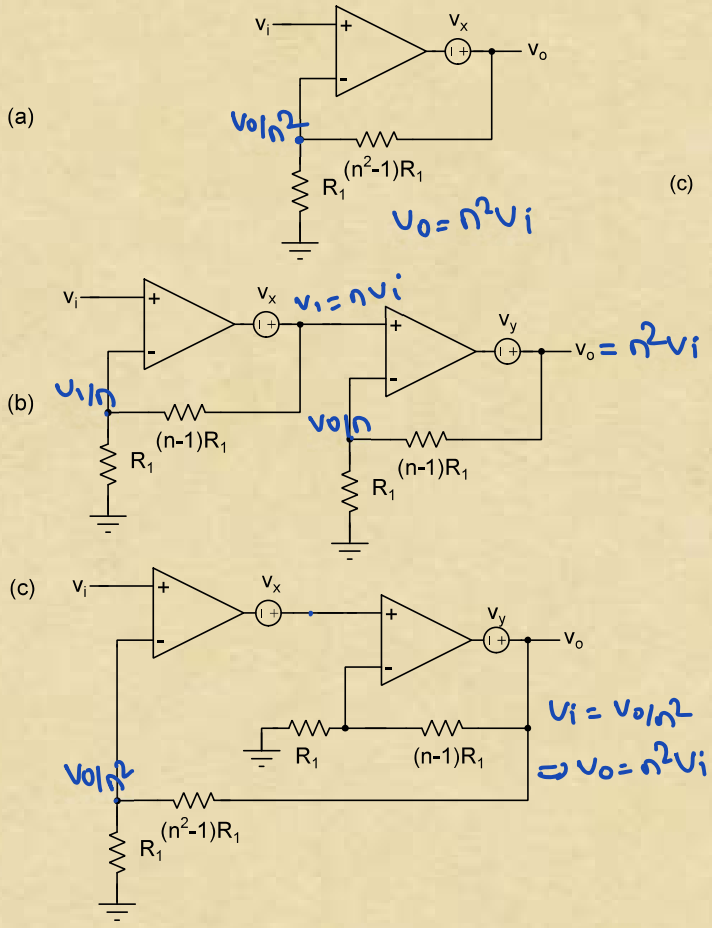


Figure 17: Circuit for Problem 17.

For each of the circuits above, the opamps are ideal. Determine v_o.

Problem 18

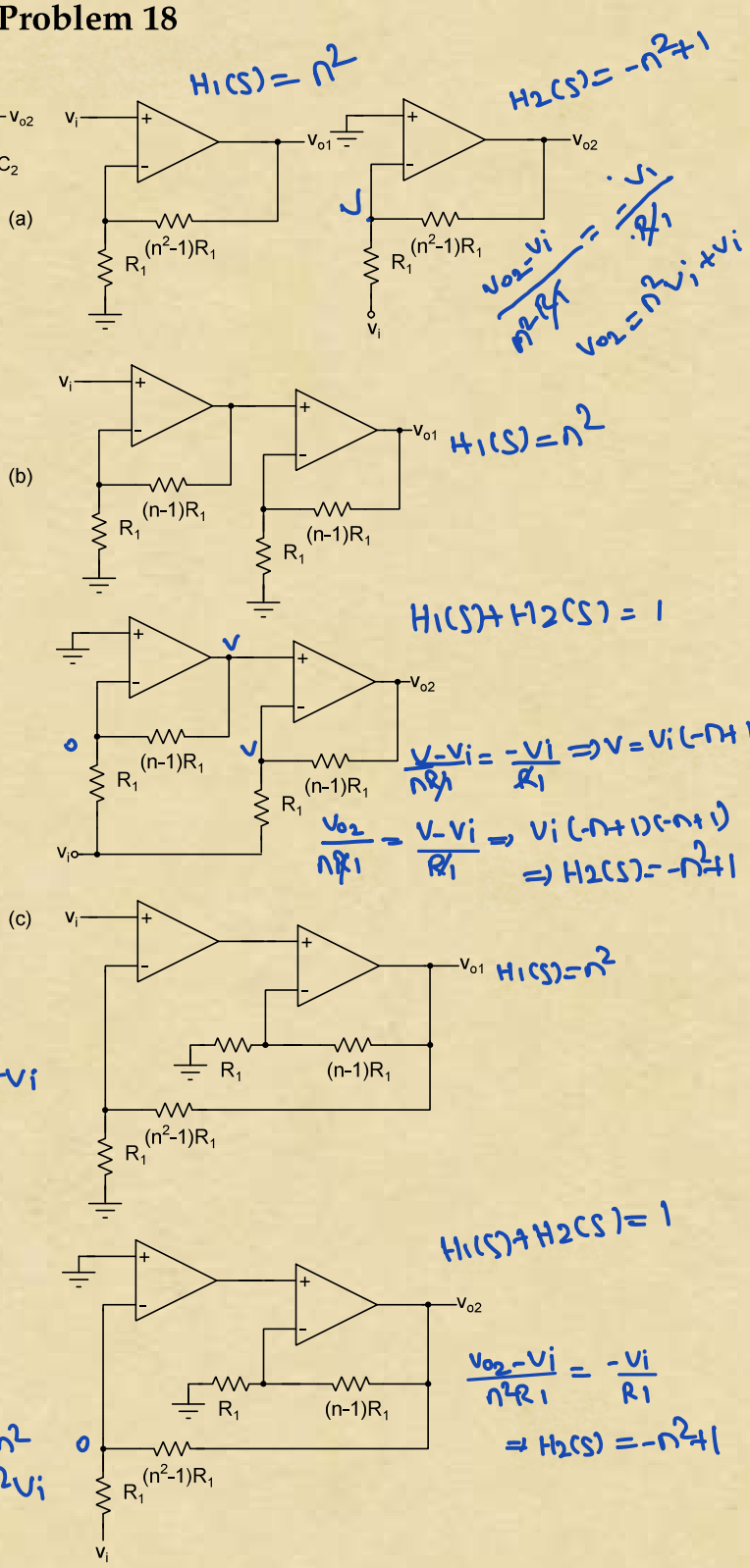


Figure 18: Circuit for Problem 18.

The opamps are ideal. For each of the networks in parts (a), (b) and (c) above, determine $H_1 = v_{o1}/v_i$ and $H_2(s) = v_{o2}/v_i$. Is there a pattern you notice?

Recall that the ideal opamp is a VCVS with infinite

gain and one of its output terminals grounded and inaccessible. How is it that the pattern of Problem 15 holds here, though not in Problem 16?

Problem 19

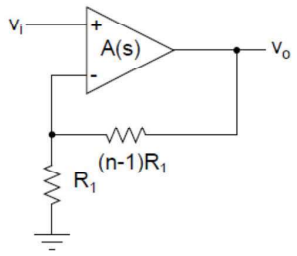


Figure 19: Circuit for Problem 19.

In class, we first assumed that the opamp was a VCVS with infinite gain. We then said – well, in reality, the gain is not infinite, but a large number. In either case, the VCVS had no "memory", meaning that it is infinitely quick; a change in the input is instantly reflected in the output.

Not surprisingly, it turns out that the opamp is not infinitely fast either. We now, therefore, need to talk about the **transfer function** $V_o(s)/V_i(s)$. This problem will explore the effect of the "slowness" of the opamp on the feedback system. Recall from your prior classes that, for a system to be stable, its poles must be in the left half s-plane.

- For $A(s) = \frac{A_o}{1+\frac{s}{\omega_o}}$, determine the closed-loop transfer function $V_o(s)/V_i(s)$. Determine the 3-dB bandwidth of the loop-gain function. Also determine the 3-dB bandwidth of the closed loop amplifier. How is it related to that of the loop-gain? Plot the locus of the poles of the closed loop system as A_o is varied from $0 - \infty$. Comment on the stability of the closed-loop system.
- Let us assume a more complicated model for $A(s)$. Let $A(s) = \frac{A_o}{(1+\frac{s}{\omega_o})^2}$. Determine the closed-loop transfer function $V_o(s)/V_i(s)$. Express the transfer function in the form
$$\frac{V_o(s)}{V_i(s)} = \frac{G}{\frac{s^2}{\omega_p^2} + \frac{s}{Q_p\omega_p} + 1}, \quad (2)$$
where G is the dc gain. Plot the locus of the poles of the closed loop system as A_o is varied from $0 - \infty$. Comment on the stability of the closed-loop system.
- Let us assume an even more complicated model for $A(s)$. Let $A(s) = \frac{A_o}{(1+\frac{s}{\omega_o})^3}$. Determine the closed-loop transfer function $V_o(s)/V_i(s)$. Plot the locus of the poles of the closed loop system as A_o is varied from $0 - \infty$. Comment on the stability of the closed-loop system.

Problem 20: LTSpice Simulation

a. Design the circuit shown in figure 2 in LTSpice using ideal op-amp and observe the output when a sinusoidal input of amplitude 1V and frequency 1kHz is applied at V1, V2 and V3. Verify that Vo=-(V1+V2+V3) when R1=R2=R3=R4.
b. Design the circuit in figure 3 with real op-amp (supply limited) observe Vo when a triangular waveform of 1kHz as show in figure is applied at Vi. Vary amplitude of the triangular wave from 1V to 5V with step of 1V and observe the output. Find the maximum amplitude which can be applied at Vi for distortion free output (Vo is not saturated).

$$V_o = \frac{A_o}{1+\frac{s}{\omega_o}}\left(V_i - \frac{V_o}{n}\right)$$

$$\left(\frac{(1+s/\omega_o)}{A_o} + \frac{1}{n}\right) V_o = V_i$$

$$\Rightarrow \frac{V_o}{V_i} = \frac{A_o n}{n(1+s/\omega_o)+A_o} = \frac{\frac{A_o n}{A_o+n}}{1+\frac{sn}{\omega_o(A_o+n)}}$$

$$\Rightarrow \omega_{3dB} = \frac{\omega_o(A_o+n)}{n}$$

LG: $\frac{A_o}{1+s/\omega_o}\cdot\frac{1}{n}$ LG: $A(s)\cdot\beta$

$$T(0) = \frac{A_o}{1+LG}$$

$$s = -(1+A_o/n)\omega_o$$

All poles $<0 \Rightarrow$ stable

$$A(s) = \frac{A_0}{(1+s/\omega_0)^2} \qquad V_0 = \frac{A_0}{(1+s/\omega_0)^2}\left(V_i - \frac{V_0}{n}\right) \Rightarrow \frac{V_0(1+s/\omega_0)^2}{A_0} + \frac{V_0}{n} = V_i$$

$$\Rightarrow \frac{V_0}{V_i} = \frac{A_0 n}{n(1+s/\omega_0)^2 + A_0} = \frac{\frac{A_0 n}{A_0+n}}{1 + \left(\frac{2/\omega_0}{A_0+n}\right)s + \left(\frac{1/\omega_0^2}{A_0+n}\right)s^2}$$

$s = \omega_0\left(\pm j\sqrt{\frac{A_0}{n} - 1}\right)$ → always left half → stable

$$A(s) = \frac{A_0}{(1+s/\omega_0)^3} \Rightarrow V_0\left(\frac{1}{n} + \frac{(1+s/\omega_0)^3}{A_0}\right) = V_i$$

$$A_0 + n(1+s/\omega_0)^3 = 0$$

$$\Rightarrow -\frac{A_0}{n} = (1+s/\omega_0)^3 \qquad \sqrt[3]{-\frac{A_0}{n}} = -t \quad t \in R$$

$$1 + s/\omega_0 = -t,\ -t\omega,\ -t\omega^2 \qquad \omega = -\frac{1}{2} + \frac{\sqrt{3}j}{2}$$

$$\Rightarrow s/\omega_0 = -t-1;\ -t\omega - 1,\ -t\omega^2 - 1 \qquad \omega^2 = -\frac{1}{2} - \frac{\sqrt{3}j}{2}$$

$-t$ → −ve

$$-\sqrt[3]{\frac{A_0}{n}}\left(-\tfrac{1}{2} + \sqrt{3}j/2\right) - 1$$

$$\frac{1}{2}\sqrt[3]{\frac{A_0}{n}} - 1$$

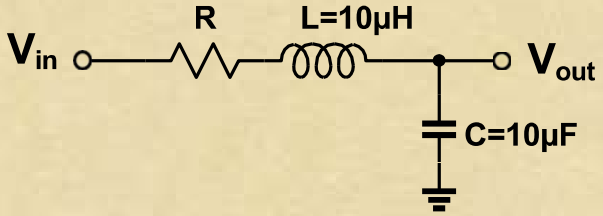
$$-\sqrt[3]{\frac{A_0}{n}}\left(-\tfrac{1}{2} - \frac{\sqrt{3}j}{2}\right) - 1$$

→ not always stable

EE2007-Analog Systems and Lab: Problem Set #2

Aniruddhan, Saurabh Saxena

1. In the RLC circuit below,



derive the transfer function $H(s)=V_{out}(s)/V_{in}(s)$ and prove that the circuit is equivalent to a standard 2nd order system with transfer function:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

a) Find the expressions for damping factor (ζ), quality factor (Q = 1/2ζ), natural frequency (w_n) and poles p_1, p_2 (roots of s) and their respective frequencies, w_{p1} and w_{p2} in terms of R, L and C.

b) Considering L=10μH and C=10μF, fill the values in the following table for the corresponding values of R.

c) Show p_1 and p_2 calculated in (b) on the s-plane and comment on the movement of poles w.r.t. damping factor (ζ).

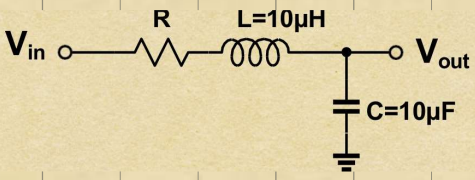
R (Ω)	ζ	Q=1/2ζ	p_1 ($\sigma + j\omega$)	p_2 ($\sigma + j\omega$)	w_{p1} (rad/s)	w_{p2} (rad/s)
0.02						
0.1						
0.4						
1						
1.4						
2						
5						
10						
20						
100						

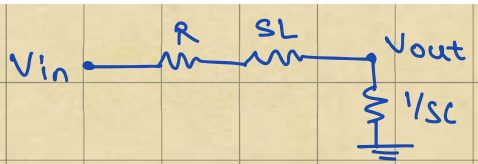
d) Enter the circuit in the LTspice and perform following simulations for all values of R given in the table:

i. Plot AC magnitude and phase response for $V_{out}(s)/V_{in}(s)$. Comment on the behavior of AC magnitude and phase response w.r.t. damping factor, ζ.

ii. Plot the transient response by applying a unit step (0 to 1V with initial delay of 1ms and T_{rise} = 1ns) for the time span of 10ms. Comment on the effect of varying ζ on the transient response.

iii. After observing the behavior of AC and transient behavior w.r.t. ζ in (i) & (ii), it is now understood that both AC and transient response are interrelated. How can you intuitively guess the approximate value of ζ by simply looking at either AC magnitude or transient response?





a)

$$\frac{V_{in}}{(R+SL+1/SC)\cdot SC} = V_{out}$$

$$\frac{V_{in}}{S^2LC+SCR+1} = V_{out} \Rightarrow H(S) = \frac{(1/LC)}{S^2+S(R/L)+(1/LC)}$$

$$\omega_n = \sqrt{1/LC} \qquad R/L = 2\xi\sqrt{1/LC} \Rightarrow \xi = \frac{R}{2}\sqrt{\frac{C}{L}} \Rightarrow Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$p_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \qquad p_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

b) $\omega_n = 10^5 \text{ rad/s} \qquad \xi = \frac{R}{2} \qquad Q = \frac{1}{R}$

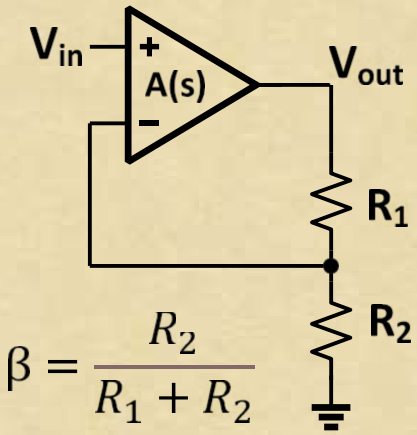
R (Ω)	ζ	Q=1/2ζ	p_1 $(\sigma + j\omega)$	p_2 $(\sigma + j\omega)$
0.02	0.01	50	$-10^3 - 9.995\times10^4 j$	$-10^3 + 9.995\times10^4 j$
0.1	0.05	10	$-5\times10^3 - 9.987\times10^4 j$	$-5\times10^3 + 9.987\times10^4 j$
0.4	0.2	2.5	$-2\times10^4 - 9.797\times10^4 j$	$-2\times10^4 + 9.797\times10^4 j$
1	0.5	1	$-5\times10^4 - 8.66\times10^4 j$	$-5\times10^4 + 8.66\times10^4 j$
1.4	0.7	0.71	$-7\times10^4 - 7.14\times10^4 j$	$-7\times10^4 + 7.14\times10^4 j$
2	1	0.5	-10^5	-10^5
5	2.5	0.2	-4.71×10^5	-20871
10	5	0.1	-9.89×10^5	-10102
20	10	0.05	-1.99×10^6	-5013
100	50	0.01	-9.99×10^6	-100

w_{p1} (rad/s)	w_{p2} (rad/s)
10^5	10^5
10^5	10^5
10^5	10^5
10^5	10^5
10^5	10^5
10^5	10^5
4.79×10^5	2.08×10^4
9.89×10^5	1.01×10^4
1.99×10^6	5×10^2
9.99×10^6	10^3

EE2007-Analog Systems and Lab: Problem Set #2

Aniruddhan, Saurabh Saxena

2. For the non-inverting amplifier shown in figure below, assuming op-amp is modelled as 2nd order transfer *A(s)*:



$$\beta = \frac{R_2}{R_1 + R_2}$$

$$A(s) = \frac{A_o}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)}$$

$$A_o = 10^4$$

a) Find the loop gain transfer function, *LG(s).*

b) Find the closed loop transfer function, $H(s)=V_{out}(s)/V_{in}(s)$, poles p_{1_cl}, p_{2_cl} (roots of s) and their respective frequencies, w_{p1_cl} and w_{p2_cl} in terms of β, A_o, w_{p1} and w_{p2}.

c) Prove that, the circuit behaves similar to the RLC circuit of problem-1 for feedback factor, β=1 (i.e. R1=0 or R2=∞) and find the expressions for damping factor (ζ), quality factor (Q=1/2ζ), natural frequency (w_n) in terms of loop gain pole frequencies, w_{p1} and w_{p2} and DC gain, A_o.

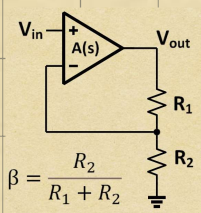
d) For $A_o=10^4$, fill values of loop gain and closed loop parameters in the following table for the corresponding w_{p1} and w_{p2}

w_{p1} (rad/s)	w_{p2} (rad/s)	Loop Gain		Closed Loop			
		w_{ugf} (rad/s)	PM (deg.)	ζ	Q=1/2 ζ	w_{p1_cl} (rad/s)	w_{p2_cl} (rad/s)
1.00E+03	1.00E+03	10^5	0	0.01	50	10^5	10^5
1.00E+02	1.00E+04	10^5	5.71°	0.05	10	10^5	10^5
2.50E+01	4.00E+04	10^5	21.8°	0.1	5	10^5	10^5
1.00E+01	1.00E+05	7.86×10^4	51.83°	0.5	1	10^5	10^5
7.07E+00	1.41E+05	6.43×10^4	65.48°	0.71	0.71	10^5	10^5
5.00E+00	2.00E+05	4.86×10^4	76.35°	1	0.5	10^5	10^5
2.00E+00	5.00E+05	2×10^4	83.91°	2.5	0.2	2.08×10^4	4.79×10^5
1.00E+00	1.00E+06	10^4	90°	5	0.1	1.01×10^4	10^6
5.00E-01	2.00E+06	5×10^3	90°	10	0.05	5.01×10^3	2×10^6
1.00E-01	1.00E+07	10^3	90°	50	0.01	10	10^7

e) Verify that increasing the spacing between loop gain poles, w_{p1} and w_{p2} has similar effect on damping factor as increasing R has in the RLC circuit of problem-1.

f) Show the locations of both loop gain poles p_1, p_2 and corresponding closed loop poles, p_{1_cl}, p_{2_cl} on s-plane for different values of ζ calculated in the above table.

g) Plot phase margin (PM) vs. ζ and find the range of ζ for which phase margin can be approximated as 100 times of ζ with +/-10% inaccuracy.



$$A(s) = \frac{A_o}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}$$

$$A_o = 10^4$$

$$\beta = \frac{R_2}{R_1+R_2}$$

a) $L.G = \beta . A(s) = \frac{A_o}{(1+s/\omega_{p1})(1+s/\omega_{p2})} \cdot \frac{R_2}{R_1+R_2}$

b) $H(s) = \frac{A(s)}{1+\beta A(s)} = \frac{\frac{A_o}{(1+s/\omega_{p1})(1+s/\omega_{p2})}}{1+\frac{A_o}{(1+s/\omega_{p1})(1+s/\omega_{p2})} \cdot \frac{R_2}{R_1+R_2}}$

$$= \frac{A_o(R_1+R_2)}{(1+s/\omega_{p1})(1+s/\omega_{p2})(R_1+R_2)+A_oR_2}$$

closed loop poles: $\frac{s^2}{\omega_{p1}\omega_{p2}} + s\left(\frac{\omega_{p1}+\omega_{p2}}{\omega_{p1}\omega_{p2}}\right) + \left(\frac{A_oR_2}{R_1+R_2}+1\right) = 0$

$$\Rightarrow s^2 + (\omega_{p1}+\omega_{p2})s + \left(\frac{A_oR_2+R_1+R_2}{R_1+R_2}\right)\omega_{p1}\omega_{p2} = 0$$

$$s = \frac{-(\omega_{p1}+\omega_{p2}) \pm \sqrt{(\omega_{p1}+\omega_{p2})^2 - 4\omega_{p1}\omega_{p2}\left(1+\frac{A_oR_2}{R_1+R_2}\right)}}{2}$$

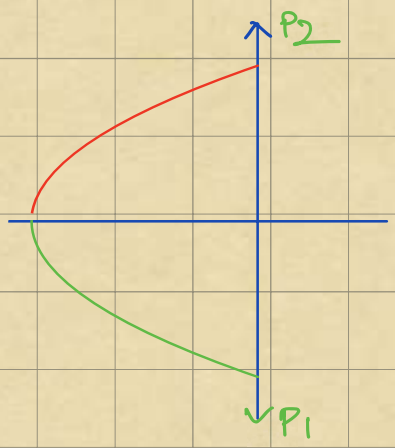
c) $R_1 = 0 \Rightarrow \omega_n^2 = \omega_{p1}\omega_{p2}(1+A_o) \Rightarrow \omega_n = \sqrt{\omega_{p1}\omega_{p2}(1+A_o)}$

$2\xi\omega_n = \omega_{p1}+\omega_{p2} \Rightarrow \xi = \frac{\omega_{p1}+\omega_{p2}}{2\sqrt{\omega_{p1}\omega_{p2}(1+A_o)}}$, $Q = \frac{\sqrt{\omega_{p1}\omega_{p2}(1+A_o)}}{\omega_{p1}+\omega_{p2}}$

d) $\left|\frac{A_o}{\sqrt{1+\frac{\omega_{ugf}^2}{\omega_{p1}^2}}\sqrt{1+\frac{\omega_{ugf}^2}{\omega_{p2}^2}}}\right| = 1$

e) in RLC increasing R increases damping factor ξ which smooths out the response. If ω_{p2} is increased very high, $\omega_{p1}+\omega_{p2}$ grows faster than the denominator eqv. root $\Rightarrow$ higher ξ prevents oscillations

f)



h) Enter the circuit in LTspice and perform following simulations for all the values of w_{p1}, w_{p2} listed in the table:

   i. Plot the AC magnitude and phase response of the loop gain transfer function for all values of w_{p1}, w_{p2} and corresponding AC magnitude and phase response of the closed loop transfer function. Comment on effect of increasing and decreasing phase margin on the closed loop AC magnitude and phase response.

   ii. Plot the step response by applying a unit step (0 to 1V with initial delay of 1ms and T_{rise} = 1ns) for the time span of 10ms. Comment on the effect of varying phase margin and ζ on the transient response. Find the phase margin and corresponding value of ζ for the fastest settling (output settles within 95% of the final value).

   iii. Change the value of feedback factor, β from 1 to 1/10 (i.e. $R_1=10R_2$) and observe the effect of increasing closed loop gain ($k=1/\beta$) on phase margin, damping factor and unity gain frequency (w_{ugf}). Comment on the result.

ANALOG SYSTEMS : PROBLEM SET 3

Problem 1

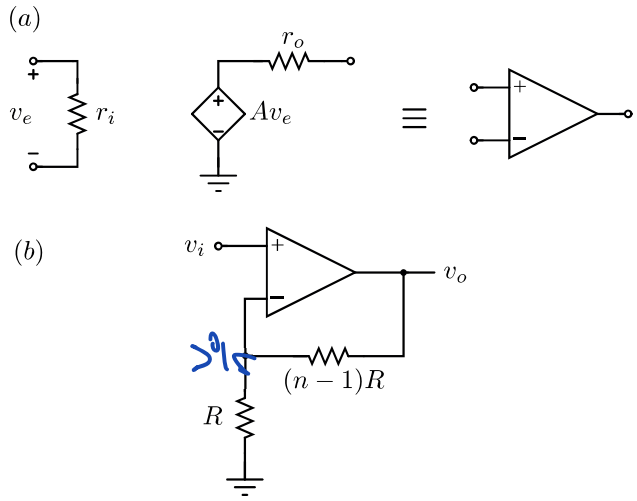


Figure 1: (a) Simplified equivalent circuit of a VCVS, with a large A. (b) The VCVS embedded in a negative feedback loop.

In class, we saw how a VCVS with a large but imprecise gain can be embedded inside a negative feedback loop to realize a VCVS with a stable gain. It turns out that in practice, the imprecise VCVS, shown in Fig. 1(a), is not all that ideal – its input resistance is finite, and its output resistance is non-zero. The symbol for the imprecise VCVS is also shown in Fig. 1(a). It is realized to make the VCVS shown in Fig. 1(b). If $A = \infty$, $v_o/v_i = n$.

Determine the input resistance, gain and output resistance of the VCVS of Fig. 1(b) in terms of A, r_i, r_o, R and n.

Evaluate the quantities above under the following limiting conditions.

a. Input resistance and gain when $r_o = 0$.

b. Output resistance and gain when $r_i = \infty$.

c. $A \to \infty$.

Problem 2

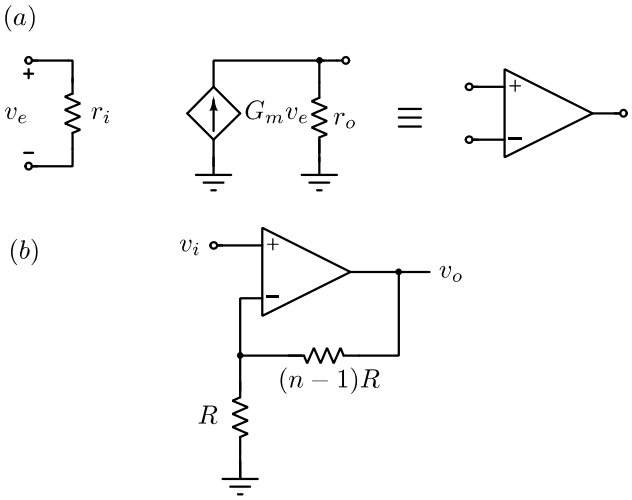


Figure 2: (a) Simplified equivalent circuit of a VCCS, with a large G_m. (b) The VCCS embedded in a negative feedback loop.

Rather than use a VCVS with a large (but uncertain) gain, this problem attempts to use a VCCS with a large (but uncertain) transconductance G_m. Further, the imprecise VCCS, shown in Fig. 2(a), is not all that ideal – its input and output resistances are finite. For simplicity, we use the same symbol for the imprecise VCCS as in Fig. 1(a). It is realized to make the VCVS shown in Fig. 2(b). What is v_o/v_i when $G_m = \infty$?

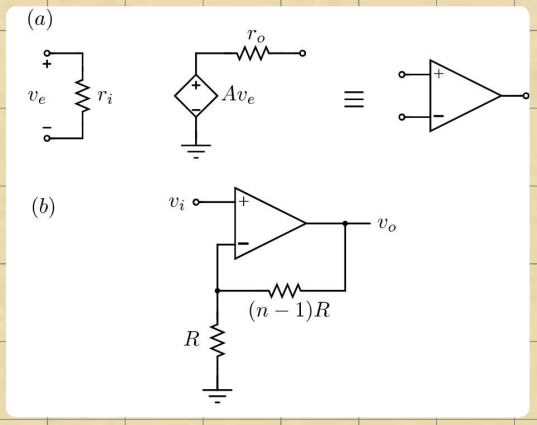
Determine the input resistance, gain and output resistance of the VCVS of Fig. 2(b) in terms of G_m, r_i, r_o, R and n.

Evaluate the quantities above under the following limiting conditions.

a. Input resistance and gain when $r_o = \infty$.

b. Output resistance and gain when $r_i = \infty$.

c. $A \to \infty$.



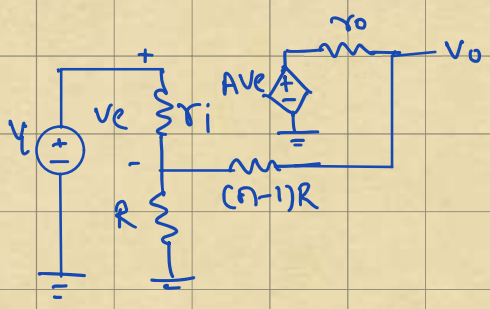
Determine the input resistance, gain and output resistance of the VCVS of Fig. 1(b) in terms of A, r_i, r_o, R and n.

Evaluate the quantities above under the following limiting conditions.

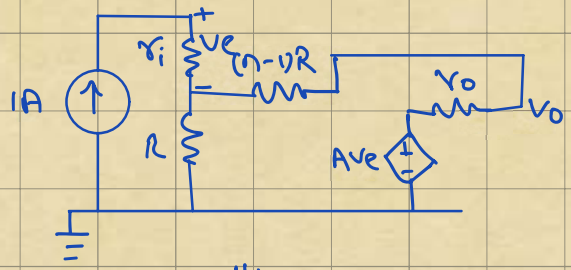
a. Input resistance and gain when $r_o = 0$.

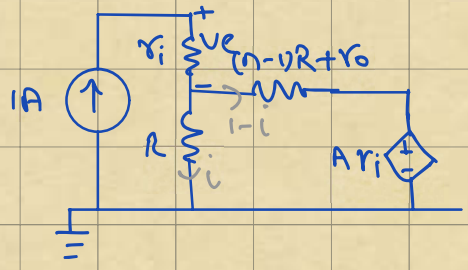
b. Output resistance and gain when $r_i = \infty$.

c. $A \to \infty$.



(a) R_{in}



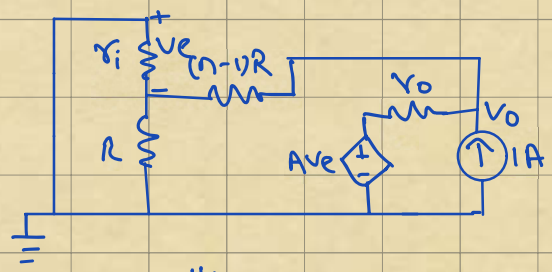


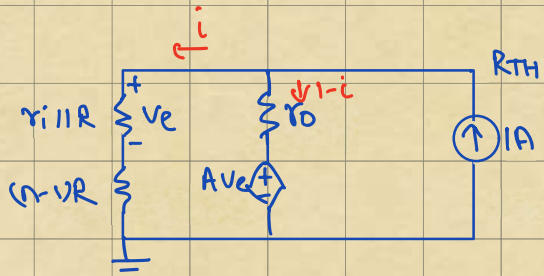
$$iR = (r_o + (n-1)R)(1-i) + Ar_i$$

$$\Rightarrow i = \frac{Ar_i + r_o + (n-1)R}{nR + r_o}$$

$$R_{in} = 1r_i + iR = r_i + R\left[\frac{Ar_i + r_o + (n-1)R}{nR + r_o}\right]$$

(b)



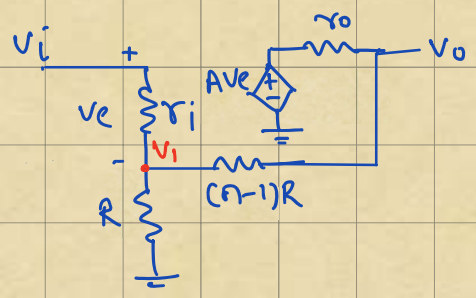


$$(1-i)r_o - \frac{A i r_i R}{r_i+R} = \frac{i r_i R}{r_i+R} + (n-1)iR$$

$$\Rightarrow i = \frac{r_o}{r_o + \frac{r_i R}{r_i+R}(1+A) + (n-1)R}$$

$$R_{out} = i\left(\frac{r_i R}{r_i+R} + (n-1)R\right) = \frac{r_o\left(\frac{r_i R}{r_i+R} + (n-1)R\right)}{r_o + \frac{r_i R}{r_i+R}(1+A) + (n-1)R}$$

(c)



$$V_e = V_i - V_1 \qquad \frac{V_e}{r_i} = \frac{V_i - V_e}{R} + \frac{V_i - V_e - V_o}{(n-1)R}$$

$$V_e\left[\frac{1}{r_i} + \frac{n}{(n-1)R}\right] = V_i\left[\frac{n}{(n-1)R}\right] - \frac{V_o}{(n-1)R}$$

$$\frac{V_o - AV_e}{r_o} = \frac{V_i - V_e - V_o}{(n-1)R} \Rightarrow V_e\left[\frac{1}{(n-1)R} - \frac{A}{r_o}\right] = \frac{V_i}{(n-1)R} - \frac{V_o}{(n-1)R} - \frac{V_o}{r_o}$$

$$\Rightarrow \frac{1/r_i + n/(n-1)R}{1/(n-1)R - A/r_o} = \frac{n/(n-1)R - \text{Gain}/(n-1)R}{1/(n-1)R - \text{Gain}/(n-1)R - \text{Gain}/r_o}$$

- $r_o \to 0 \Rightarrow$ $R_{in} = r_i(1 + A/n) + \left(\frac{n-1}{n}\right)R$

  $Gain = \frac{Ar_i}{\left(\frac{n-1}{n}\right)R + r_i(1 + A/n)}$

- $r_i \to \infty \Rightarrow$ $R_{out} = \frac{1}{(1 + A/n)(1/r_o) + 1/(nR)}$

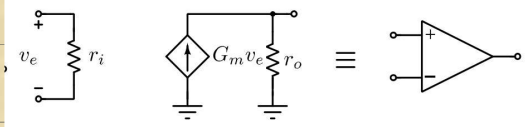
  $Gain = \frac{1}{\frac{r_o}{nAR} + 1/A + 1/n}$

- $A \to \infty \Rightarrow$ $R_{in} = \infty$

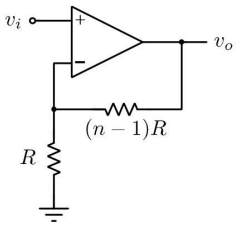
  $R_{out} = 0$

  $Gain = n$

(a)



(b)



to make the VCVS shown in Fig. 2(b). What is v_o/v_i when $G_m = \infty$?

Determine the input resistance, gain and output resistance of the VCVS of Fig. 2(b) in terms of G_m, r_i, r_o, R and n.

Evaluate the quantities above under the following limiting conditions.

a. Input resistance and gain when $r_o = \infty$.

b. Output resistance and gain when $r_i = \infty$.

c. $A \to \infty$.

same but $A = G_m r_o$ by source transformation

- $r_o = \infty \Rightarrow$ $R_{in} = r_i + R + G_m r_i R$

  $Gain = \frac{nr_i(G_m R(n-1) - 1) + R(n-1) + nr_i}{(n-1)R + nr_i + r_i(G_m R(n-1) - 1)}$

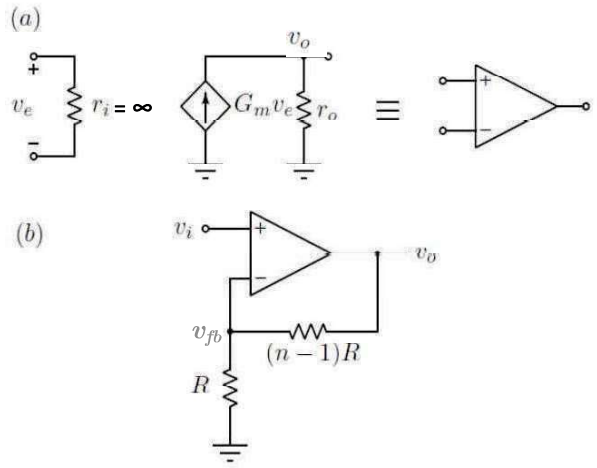
- $r_i = \infty \Rightarrow$ $R_{out} = \frac{1}{\frac{1}{r_o} + \frac{1}{nR} + \frac{G_m}{n}}$

  $Gain = \frac{nr_o(G_m R(n-1) + 1) + nr_o}{n[(n-1)R + r_o] + r_o(G_m R(n-1) - 1)}$

- $A \to \infty \Rightarrow$ $R_{in} = \infty$ $\quad Gain = n$

  $R_{out} = 0$

Problem 3



Design the circuit shown in figure (a) and (b) in LTSpice and perform following simulations:

a. Configure the VCVS shown in figure (a) for gain, A_o=5 (G_m=1mS, r_o=5k) and A_o=10 (G_m=5mS r_o=2kΩ). Give a sinusoidal input of 1V amplitude at input v_e, observe the output voltage across *ro*) and verify that gain (v_o/v_e) is same as $G_m r_o$.
b. Connect a load resistance R_L=1kΩ at v_o (in parallel with r_o) and repeat exercise in a. Comment on the effect of R_L on the gain.
c. Configure the VCVS of figure (a) in negative feedback as shown in figure (b) with following parameters and analyze the gain (v_o/v_i) by giving sinusoidal input of 1V amplitude at input v_i and observe the feedback voltage v_{fb} and output v_o. Compare the results with analysis performed in a. and b. Determine the effect of VCVS gain (A_o=$G_m r_o$) and *n* on the closed loop gain (v_o/v_i).
    i. G_m=2mS, *ro*=10k, *R*=1MΩ, *n*=2
    ii. G_m=2mS, *ro*=50k, *R*=1MΩ, *n*=5
    iii. G_m=1mS, *ro*=200k, *R*=1MΩ, *n*=2
    iv. G_m=1mS, *ro*=100k, *R*=1MΩ, *n*=10
    v. G_m=10mS, *ro*=100k, *R*=1MΩ, *n*=10

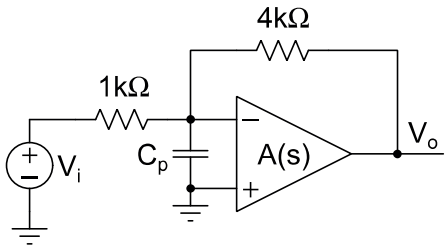


Figure 4: Circuit for problem 4

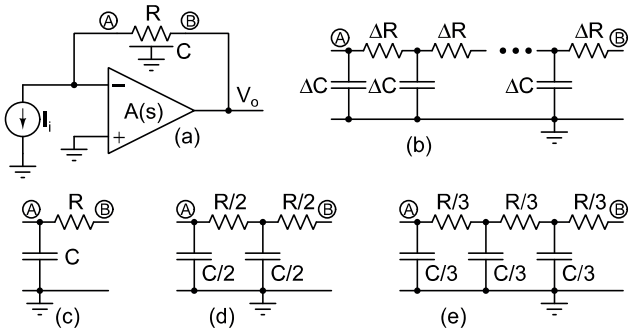


Figure 5: Circuit for problem 5

Problem 4

Fig. 4 shows an inverting amplifier. The opamp has a gain $A(s) = A_0/(1 + s/p_1)(1 + s/p_2)$ where $A_0 = 20000$, $p_1 = 1$ krad/s, $p_2 = 10$ Mrad/s. C_p is a parasitic capacitor.

a) What is the phase margin of the system with $C_p = 0$?

b) What is the closed loop bandwidth of the system? (Calculate this from (a) Unity loop gain frequency, (b) Natural frequency of the second order system, and (c) Exact calculation-computing the frequency at which the gain magnitude drops to $1/\sqrt{2}$ times the dc gain.; Compare the estimates so obtained)

c) What is the value of C_p for which the circuit becomes unstable?

d) With C_p being the value calculated in the previous part, can you change the circuit so that the phase margin is 60° without changing the opamp or the closed-loop dc gain V_o/V_i?

Problem 5

Fig. 5 shows a transimpedance amplifier. The opamp has a frequency independent gain A_0. The feedback resistor R has a parasitic capacitor C. C is distributed across the length of the resistor and should be modeled as shown in Fig. 2(b) where the infinite number of infinitesimal ΔR and ΔC sum up to R and C respectively. This cannot be analyzed easily, so we model it as shown in Fig. 2(c), (d), or (e). Analyze each case and comment on the effect of A_0 on stability or damping. (In addition to stability, this problem also tells you something about oversimplified models).

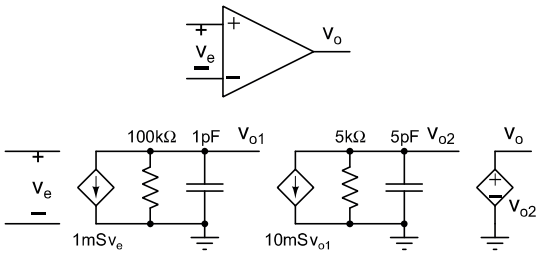


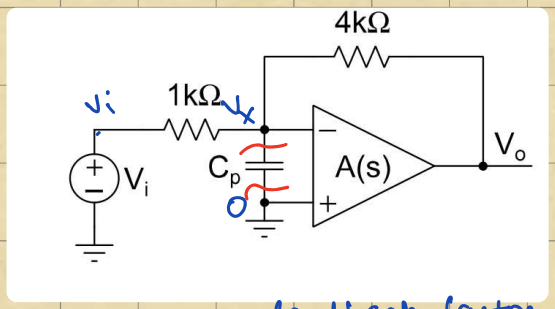
Figure 6: Circuit for problem 6

Problem 6

Fig. 6 shows the internal schematic of an opamp. This opamp is used to realize a unity gain, non-inverting amplifier.

- What is the phase margin?
- Connect a capacitor across one of the existing capacitors inside the opamp so that the phase margin is 60°.

Repeat the above if the opamp is used to realize an inverting amplifier of gain -4.



$A(s) = A_0/(1 + s/p_1)(1 + s/p_2)$

$A_0 = 20000,\ p_1 = 1\,\text{krad/s},\ p_2 = 10\,\text{Mrad/s}$

feedback factor

a) $C_p = 0 \Rightarrow \beta = \frac{10^3}{10^3 + 4\times10^3} = 0.2 \Rightarrow L.G = A(s)\cdot\beta = \frac{4000}{\left(1+\frac{s}{10^3}\right)\left(1+\frac{s}{10^7}\right)}$

$\omega_{ug} = 4\times10^6$ rad/s $\Rightarrow PM = 180 - \tan^{-1}\left(\frac{4\times10^6}{10^3}\right) - \tan^{-1}\left(\frac{4\times10^6}{10^7}\right)$

$\approx 68.2°$

b) $\omega_{ug} = 4\times10^6$ rad/s

$\omega_n = \sqrt{p_1 p_2 (1 + A_0\beta)} = 6.32\times10^6$ rad/s

$V_x = \frac{V_o + 4V_i}{5}$ $\quad V_o = -AV_x \Rightarrow \frac{V_o}{V_i} = \frac{-4}{1+s/A}$

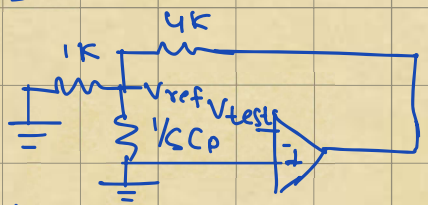
$H = \frac{-8\times10^4}{2\times10^4 + 5\left(1+\frac{s}{10^3}\right)\left(1+\frac{s}{10^7}\right)} = \frac{-16\times10^{13}}{s^2 + 10^7 s + 4\times10^{13}}$

$\omega_n^2 = 4\times10^{13} \Rightarrow \omega_n = 6.32\times10^6$ rad/s

|DC gain| = 4 $\Rightarrow$ Gain at CLBW = $2\sqrt{2}$

$2\sqrt{2} = \frac{16\times10^{13}}{\sqrt{(4\times10^{13} - \omega^2)^2 + 10^4\omega^2}} \Rightarrow \omega = 5.59\times10^6$ rad/s

c) To find LCS)



$V_{ref} = \frac{V_o\,(1K \,||\, 1/sC_p)}{4K + (1K \,||\, 1/sC_p)}$ $\quad V_o = -A(s)V_{test}$

$$\frac{V_{ref}}{V_{test}} = L(s) = \frac{A(s)}{1 + 4(1+10^3 sC_p)} = \frac{A(s)}{5 + 4\times10^3 sC_p}$$

$$L(s) = \frac{4\times10^3}{(1+s/10^3)(1+s/10^9)(1+s/(1/800C_p))}$$

$$\frac{4\times10^3}{(1+s/10^3)(1+s/10^7)(1+s/p_3)} \Rightarrow \frac{16\times10^6}{(1+\omega^2/10^6)(1+\omega^2/10^{14})(1+\omega^2/p_3^2)} = 1$$

$$\omega^2 = 16\times10^{12} \Rightarrow \omega = 8\times10^6$$

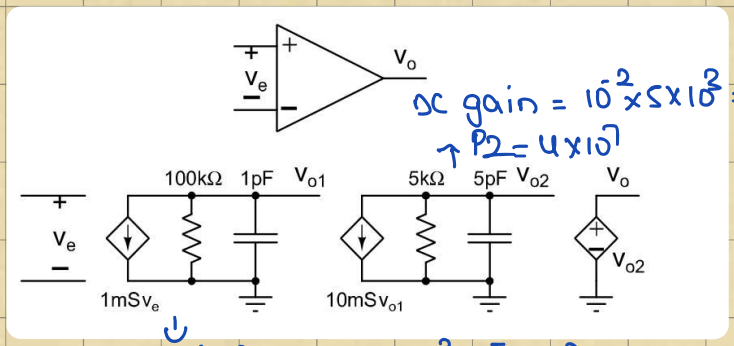
$$180 - \tan^{-1}\left(\frac{8\times10^6}{10^3}\right) - \tan^{-1}\left(\frac{8\times10^6}{10^7}\right) - \tan^{-1}\left(\frac{8\times10^6}{p_3}\right) < 0$$

$$\Rightarrow 90 - 38.66^\circ - \tan^{-1}\left(\frac{8\times10^6}{p_3}\right) < 0$$

$$\Rightarrow \tan^{-1}\left(\frac{8\times10^6}{p_3}\right) > 51.34 \Rightarrow \frac{8\times10^6}{p_3} = 1.25$$

$$p_3 = 64\times10^5 = \frac{1}{800C_p} \Rightarrow C_p > 2\times10^{-10}$$

d) Reduce 1K and 4K maintaining 1:4 ratio



DC gain $= 10^2 \times 5\times10^3 = 50$

$p_2 = 4\times10^7$

$$A(s) = \frac{5\times10^3}{(1+s/10^3)(1+s/4\times10^7)}$$

DC gain $A = g_m R_1 = 10^{-3}\times10^5 = 10^2$

$p_1 : 1/R_1C_1 = 10^3$ rad/s

$L(s) = A(s)\cdot\beta = A(s)$

$|L(s)| = 1 \Rightarrow \omega = 2\times10^9$

$PM = 2.03^\circ$

Cnew across C_1 to move ω_{p_1} to lower freq

$\tan^{-1}\left(\frac{\omega_{ug}}{p_2}\right) = 30^\circ \Rightarrow \omega_{ug} = 2.3\times10^7$ rad/

$|L(s=\omega_{ug})| = 1 \Rightarrow (p_1)_{new} = \frac{\omega_{ug}}{A_0} = 4600$

$C_1 = \frac{1}{R(p_1)_{new}} = \frac{1}{10^5 \times 4600} = 2.17$ nF

add 2.17 nF across existing 1nF capacitor

Gain 4: $\beta = 1/5 \Rightarrow L(s) = \frac{10^3}{(1+s/10^3)(1+s/4\times10^3)}$

$\omega_{ug} = 2\sqrt{10}\times10^3$ rad/s $\quad \phi_m = 4.52^\circ$

$\phi_m = 60 = 90 - \tan^{-1}\left(\frac{1}{4\times10^3 C}\right) \Rightarrow C = 0.433$ nF

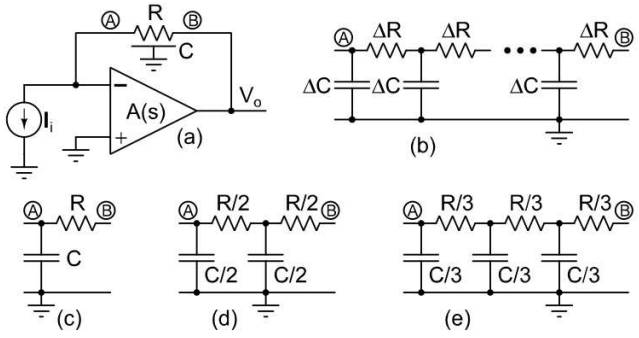
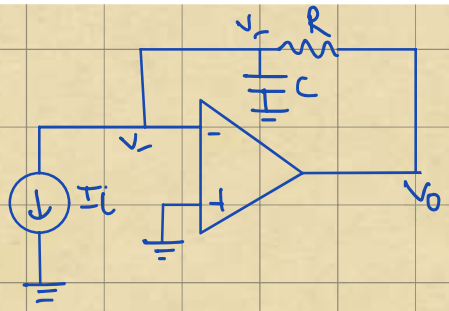


Fig. 5 shows a transimpedance amplifier. The opamp has a frequency independent gain A_0. The feedback resistor R has a parasitic capacitor C. C is distributed across the length of the resistor and should be modeled as shown in Fig. 2(b) where the infinite number of infinitesimal ΔR and ΔC sum up to R and C respectively. This cannot be analyzed easily, so we model it as shown in Fig. 2(c), (d), or (e). Analyze each case and comment on the effect of A_0 on stability or damping. (In addition to stability, this problem also tells you something about oversimplified models).



$I_i + V_-(sC) + \frac{(V_- - V_o)}{R} = 0$

$V_o = -AV_- \quad I_i + V_-(sC) + \left(\frac{V_- + AV_-}{R}\right) = 0$

$V_-\left(sC + 1/R + A/R\right) = -I_i$